Tractor ECU RE:

When a "Noise-Triggered" Recall is Also a Security Patch

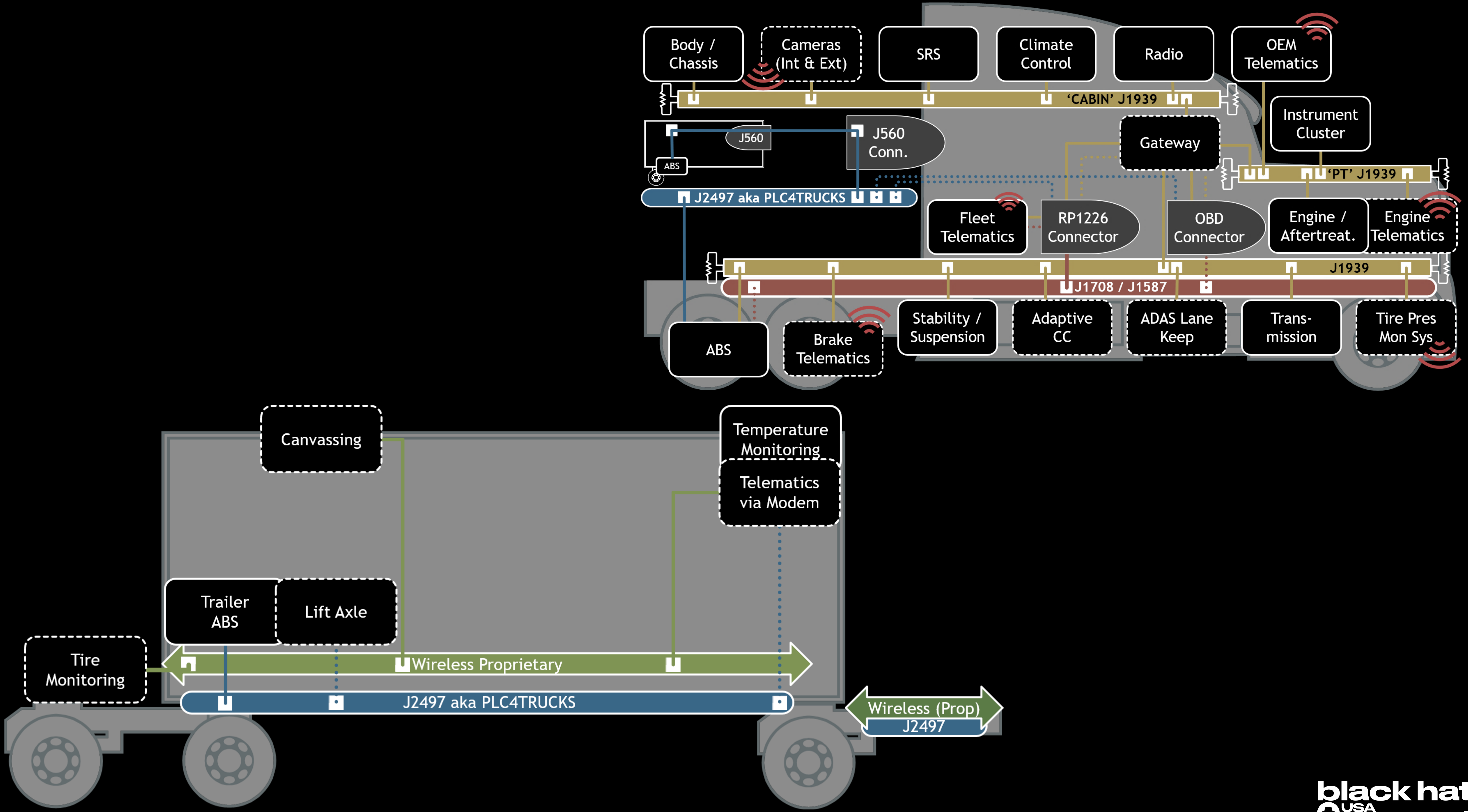
Safety-Impacting vulns patched in the recall of ~450,000 Trucks via SAE J2497 (PLC4TRUCKS)

Ben Gardiner
NMFTA Inc.

black hat USA 2026

Almost a Decade Watching Truck Cybersecurity

The National Motor Freight Traffic Assoc. has been analysing truck cybersecurity and sharing its findings since ~2017.


Body / Chassis
Cameras (Int & Ext)
SRS
Climate Control
Radio
OEM Telematics
'CABIN' J1939
J560
ABS
J560 Conn.
Gateway
Instrument Cluster
'PT' J1939
J2497 aka PLC4TRUCKS
Fleet Telematics
RP1226 Connector
OBD Connector
Engine / Aftertreat.
Engine Telematics
J1939
J1708 / J1587
ABS
Brake Telematics
Stability / Suspension
Adaptive CC
ADAS Lane Keep
Trans- mission
Tire Pres Mon Sys
Canvassing
Temperature Monitoring
Telematics via Modem
Trailer ABS
Lift Axle
Tire Monitoring
Wireless Proprietary
J2497 aka PLC4TRUCKS
Wireless (Prop)
J2497

SECTION: MOTIVATIONS

Why NMFTA chased a brake-controller recall.

Our J2497 Track Record

- Studying **J2497** (PLC4TRUCKS) protocol since 2019
- Three CVEs across J2497 and trailer equipment
- Three CISA ICS Advisories also (one for each)
- (several) Public domain technical mitigations against wireless attacks released

*PLC: Power-Line Carrier

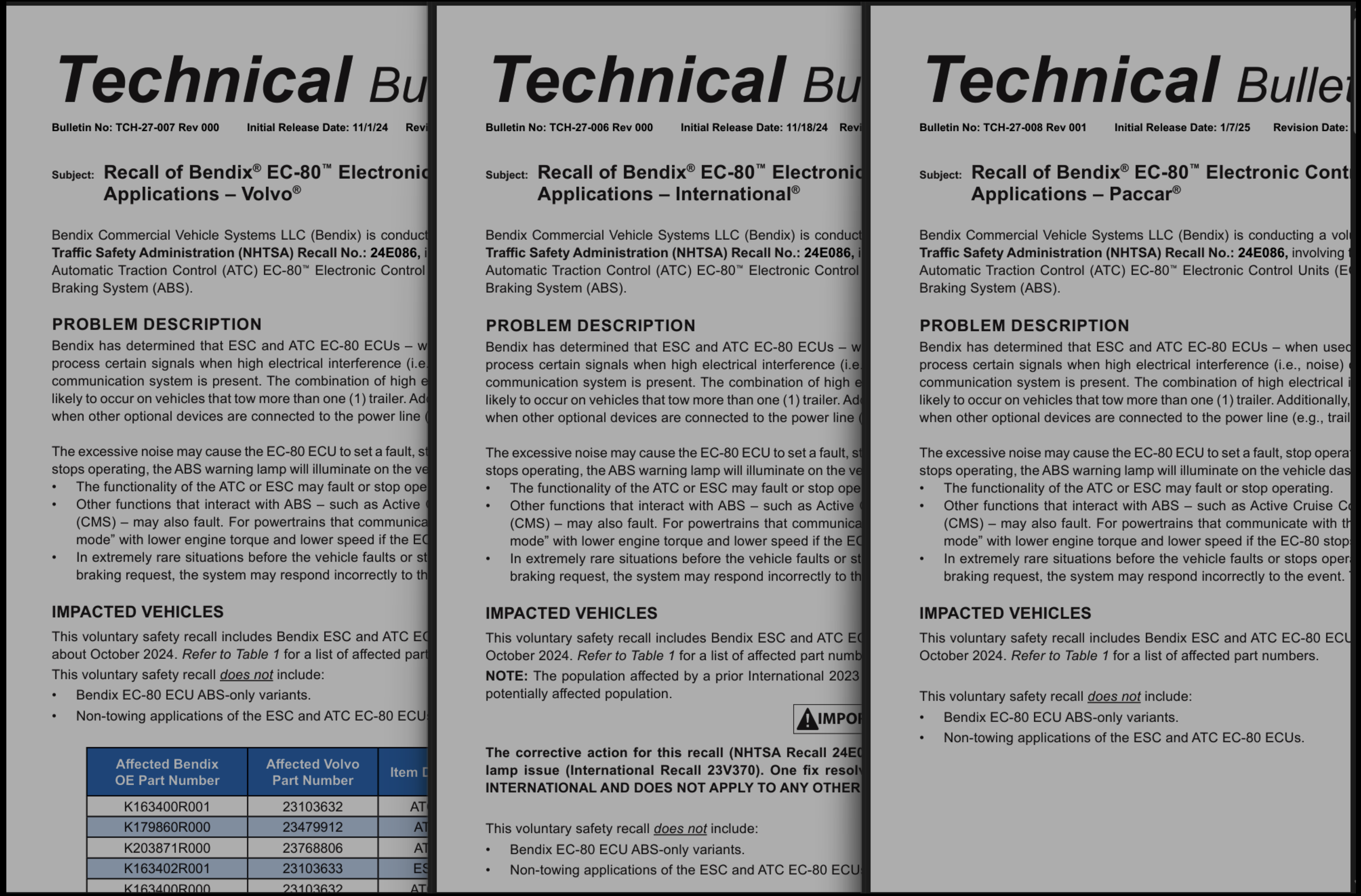
Yet we were told, repeatedly, that tractors are unaffected.

A Tip About a Recall

- An engineer friend flagged a safety recall.
- It blamed a brake-controller crash on noise in-band with the **J2497** protocol.

- Could that crash be triggered **intentionally**?

*Recall of: the Bendix EC80 Anti-lock Braking System (ABS) / Electronic Stability Program (ESP) controller.



Technical Bu

Bulletin No: TCH-27-007 Rev 000 Initial Release Date: 11/1/24 Revi

Subject: Recall of Bendix® EC-80™ Electronic Applications – Volvo®

Bendix Commercial Vehicle Systems LLC (Bendix) is conduct Traffic Safety Administration (NHTSA) Recall No.: 24E086, i Automatic Traction Control (ATC) EC-80™ Electronic Control Braking System (ABS).

PROBLEM DESCRIPTION

Bendix has determined that ESC and ATC EC-80 ECUs – w process certain signals when high electrical interference (i.e communication system is present. The combination of high e likely to occur on vehicles that tow more than one (1) trailer. Ad when other optional devices are connected to the power line (

The excessive noise may cause the EC-80 ECU to set a fault, st stops operating, the ABS warning lamp will illuminate on the ve

- The functionality of the ATC or ESC may fault or stop ope
- Other functions that interact with ABS – such as Active (CMS) – may also fault. For powertrains that communica mode" with lower engine torque and lower speed if the EC
- In extremely rare situations before the vehicle faults or st braking request, the system may respond incorrectly to th

IMPACTED VEHICLES

This voluntary safety recall includes Bendix ESC and ATC EC about October 2024. *Refer to Table 1* for a list of affected part

This voluntary safety recall *does not* include:

- Bendix EC-80 ECU ABS-only variants.
- Non-towing applications of the ESC and ATC EC-80 ECU

Affected Bendix OE Part Number	Affected Volvo Part Number	Item D
K163400R001	23103632	AT
K179860R000	23479912	AT
K203871R000	23768806	AT
K163402R001	23103633	ES
K163400R000	23103632	AT

Technical Bu

Bulletin No: TCH-27-006 Rev 000 Initial Release Date: 11/18/24 Revi

Subject: Recall of Bendix® EC-80™ Electronic Applications – International®

Bendix Commercial Vehicle Systems LLC (Bendix) is conduct Traffic Safety Administration (NHTSA) Recall No.: 24E086, i Automatic Traction Control (ATC) EC-80™ Electronic Control Braking System (ABS).

PROBLEM DESCRIPTION

Bendix has determined that ESC and ATC EC-80 ECUs – w process certain signals when high electrical interference (i.e communication system is present. The combination of high e likely to occur on vehicles that tow more than one (1) trailer. Ad when other optional devices are connected to the power line (

The excessive noise may cause the EC-80 ECU to set a fault, st stops operating, the ABS warning lamp will illuminate on the ve

- The functionality of the ATC or ESC may fault or stop ope
- Other functions that interact with ABS – such as Active (CMS) – may also fault. For powertrains that communica mode" with lower engine torque and lower speed if the EC
- In extremely rare situations before the vehicle faults or st braking request, the system may respond incorrectly to th

IMPACTED VEHICLES

This voluntary safety recall includes Bendix ESC and ATC E October 2024. *Refer to Table 1* for a list of affected part numb

NOTE: The population affected by a prior International 2023 potentially affected population.

IMPOR

The corrective action for this recall (NHTSA Recall 24E lamp issue (International Recall 23V370). One fix resol INTERNATIONAL AND DOES NOT APPLY TO ANY OTHER

This voluntary safety recall *does not* include:

- Bendix EC-80 ECU ABS-only variants.
- Non-towing applications of the ESC and ATC EC-80 ECU

Technical Bullet

Bulletin No: TCH-27-008 Rev 001 Initial Release Date: 1/7/25 Revision Date:

Subject: Recall of Bendix® EC-80™ Electronic Cont Applications – Paccar®

Bendix Commercial Vehicle Systems LLC (Bendix) is conducting a vol Traffic Safety Administration (NHTSA) Recall No.: 24E086, involving Automatic Traction Control (ATC) EC-80™ Electronic Control Units (E Braking System (ABS).

PROBLEM DESCRIPTION

Bendix has determined that ESC and ATC EC-80 ECUs – when used process certain signals when high electrical interference (i.e., noise) communication system is present. The combination of high electrical likely to occur on vehicles that tow more than one (1) trailer. Additionally, when other optional devices are connected to the power line (e.g., trail

The excessive noise may cause the EC-80 ECU to set a fault, stop opera stops operating, the ABS warning lamp will illuminate on the vehicle das

- The functionality of the ATC or ESC may fault or stop operating.
- Other functions that interact with ABS – such as Active Cruise C (CMS) – may also fault. For powertrains that communicate with th mode" with lower engine torque and lower speed if the EC-80 stop
- In extremely rare situations before the vehicle faults or stops oper braking request, the system may respond incorrectly to the event.

IMPACTED VEHICLES

This voluntary safety recall includes Bendix ESC and ATC EC-80 ECU October 2024. *Refer to Table 1* for a list of affected part numbers.

This voluntary safety recall *does not* include:

- Bendix EC-80 ECU ABS-only variants.
- Non-towing applications of the ESC and ATC EC-80 ECUs.

Current Threat Model

Fleets assume tractor safety systems are isolated from the trailer's unauthenticated PLC4TRUCKS (J2497)

The operators of the equipment believe they can ignore the J2497 attack path found in every class 8 towing application trucks since 2001.

Safety & Security (yet again)

IT

Stays **secure** if updated frequently

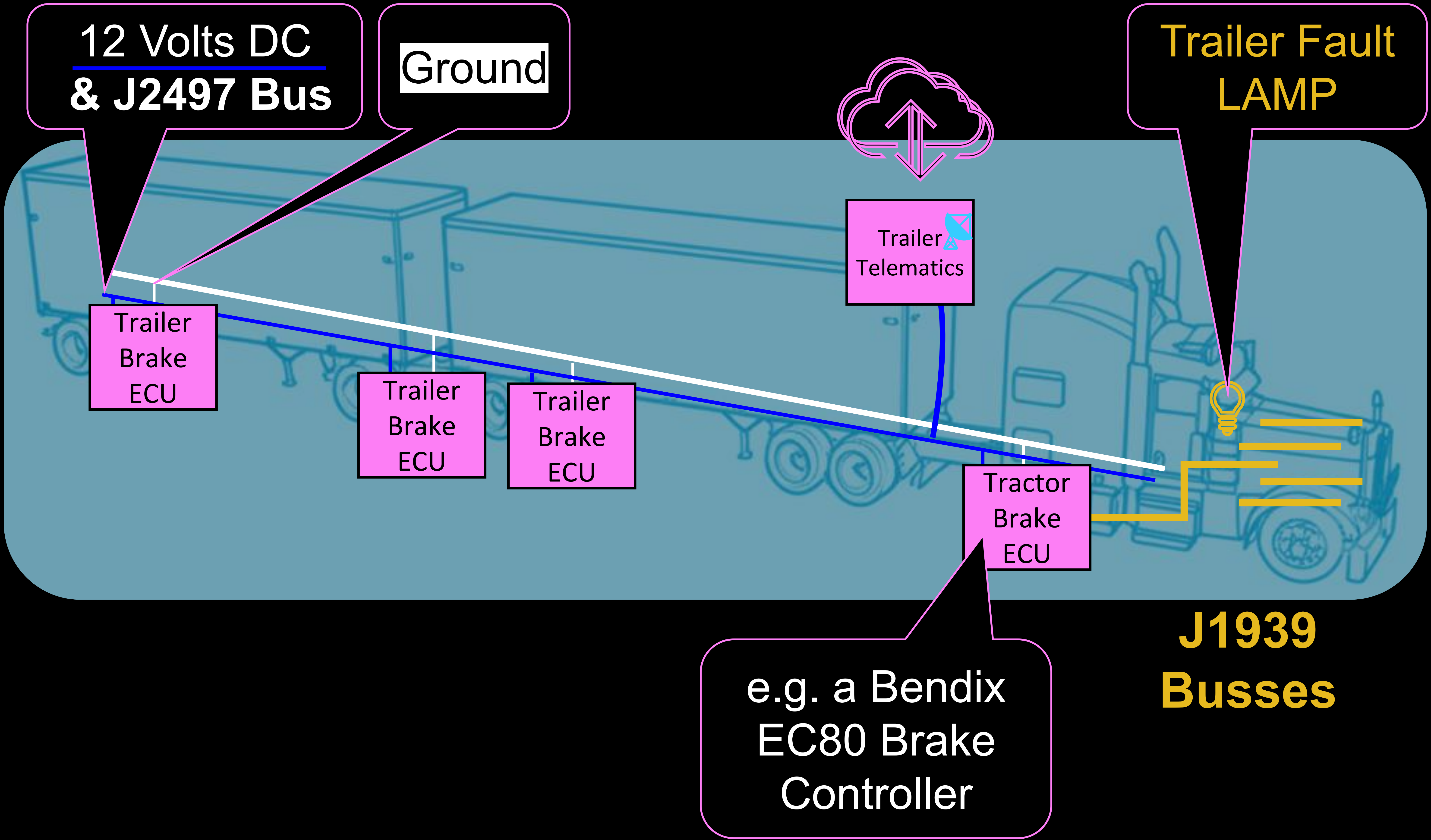
Cyber-Physical Systems

Stays **safe** if left untouched

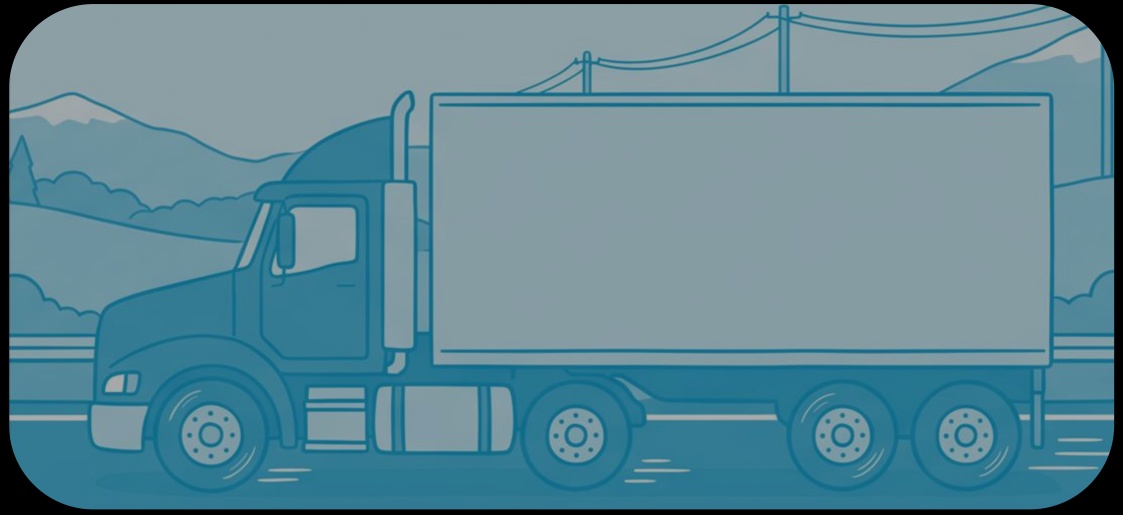
SECTION
HEAVY VEHICLES: J1939 & PLC4TRUCKS

Meet the vehicle architecture, the buses.

Trailers and Tractors



Truck Attack Surface: Diagnostics-time vs Mission-time Communications



DIAGNOSTICS	MISSION-TIME
This communication is rejected while in motion	**Works (as intended) in motion**
Authentication & Authorization: Seed-Key exchange	OLD: network access only, NEW: message authentication codes on the bus
Cyber-physical impacts. Commands, reconfigurations and also firmware dump, update	Cyber-physical impacts. Many are a de-rate.
Connections: e.g. UDS, KWP2000, XCP	Signals encoded in: SPNs (J1939), **PIDs (J1587)**
Same data busses: J1939 and J2947/J1708/J1587	

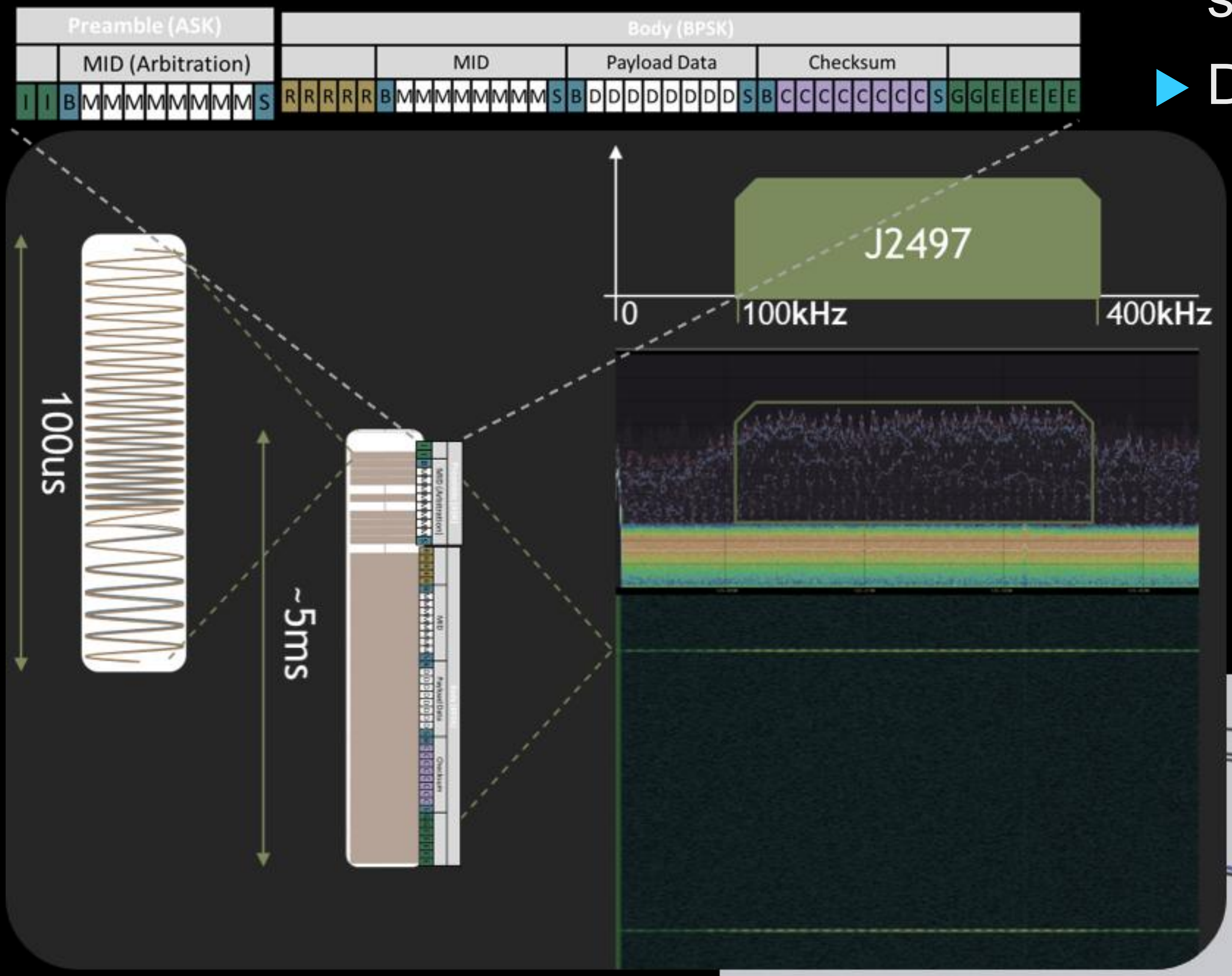
Two Buses: J2497 & J1939

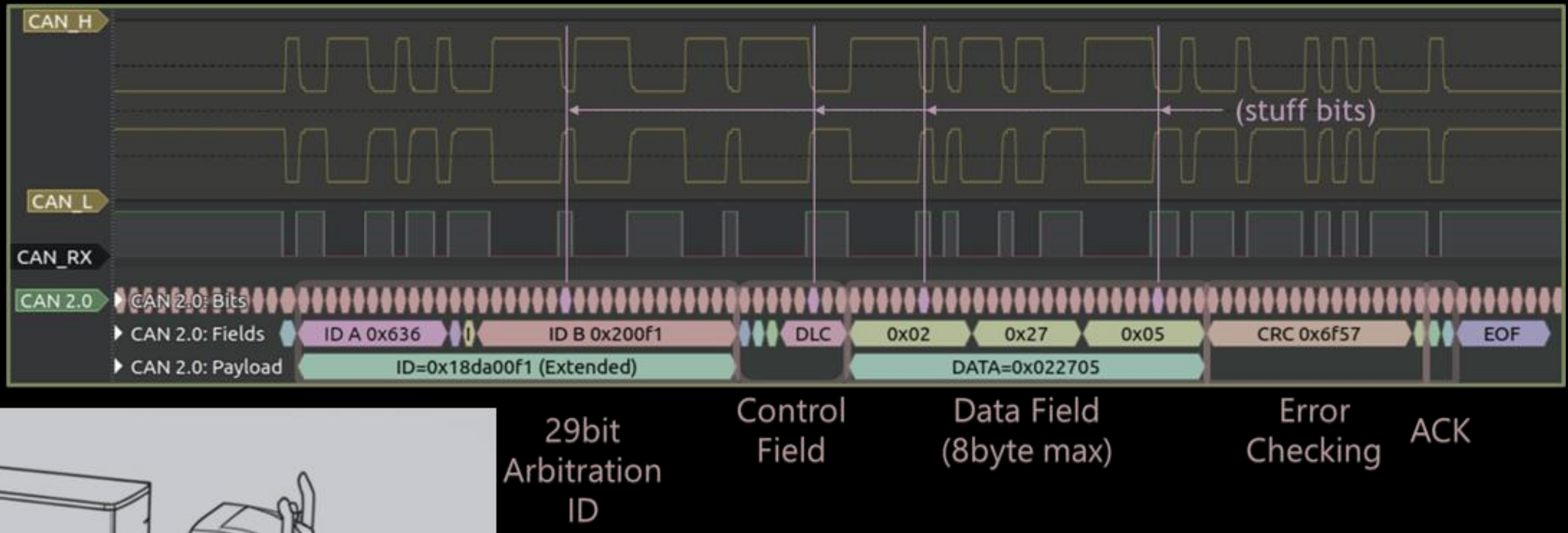
J2497 (powerline PHY for J1587)

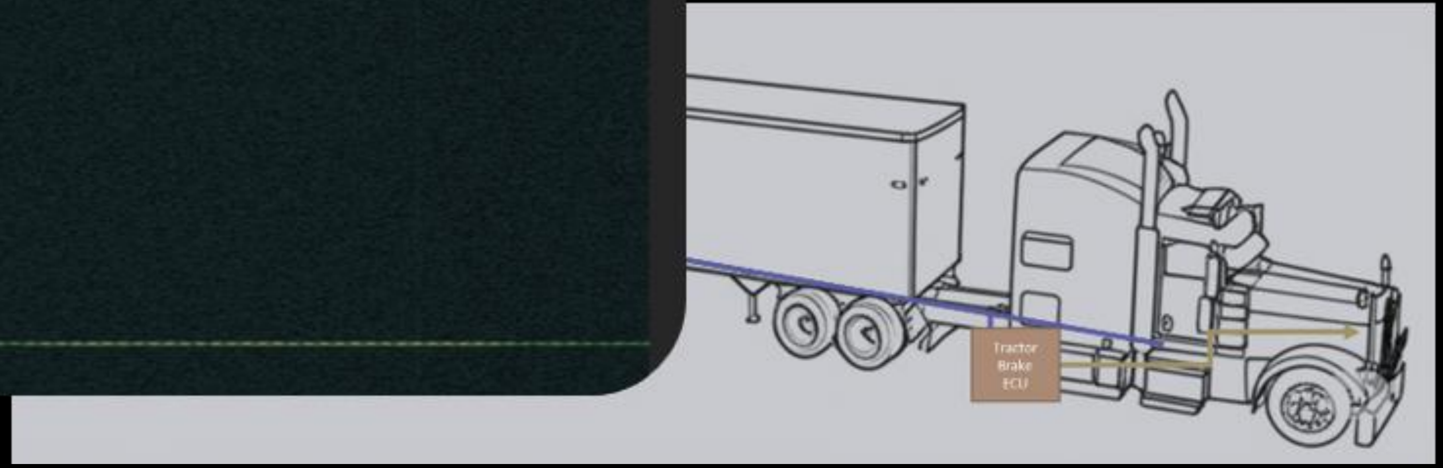
- 9600 bps
- One segment
- Mission-time LAMP and 'ABS event active' messages standardized
- Diagnostics: none

J1939 (aka CAN bus)

- 250 kbps / 500 kbps
- **Multiple segments**
- Mission-time standardized J1939 signals (and proprietary messages)
- Diagnostics both J1939-specific and UDS

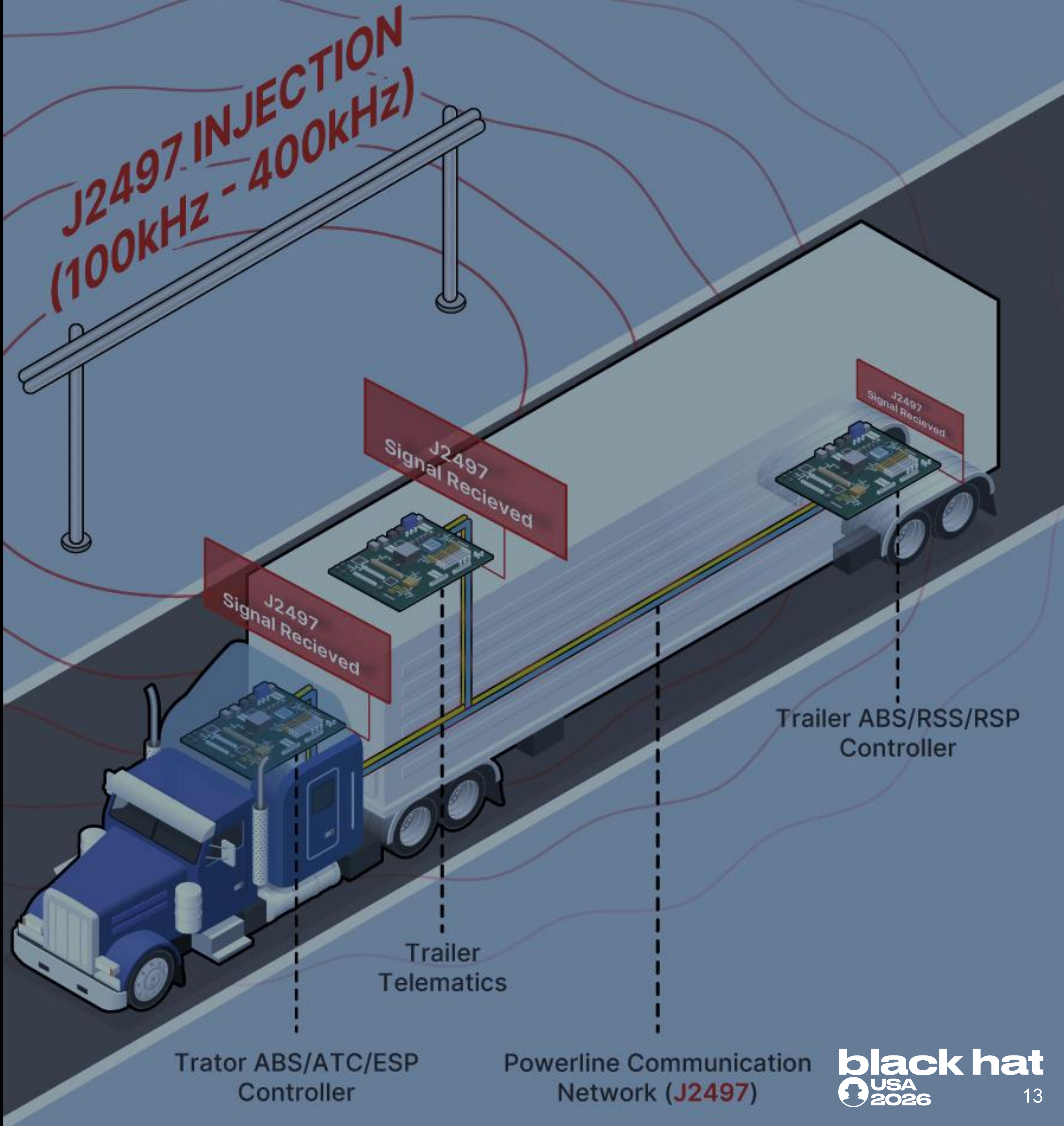




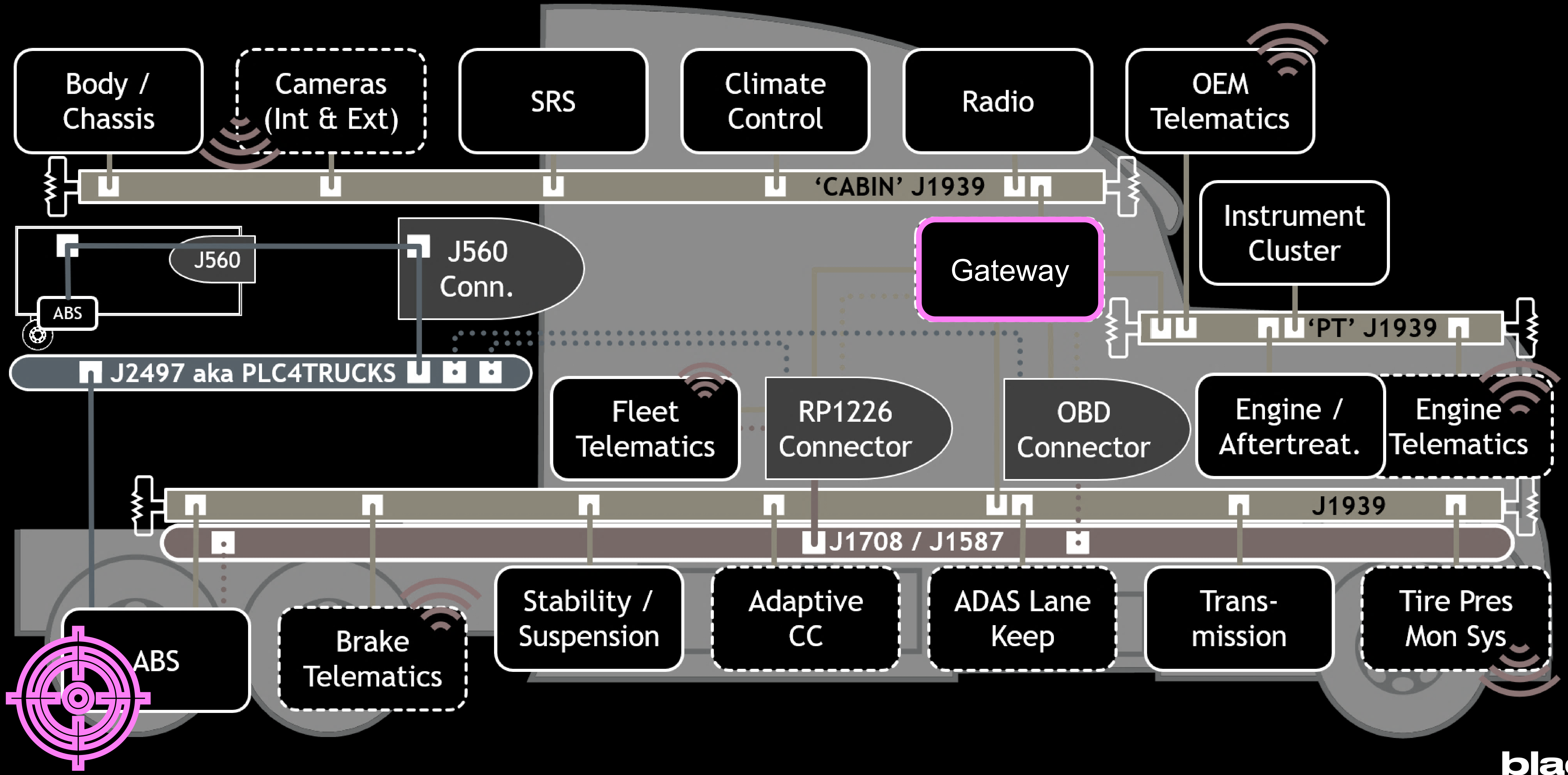


Wirelessly Accessible

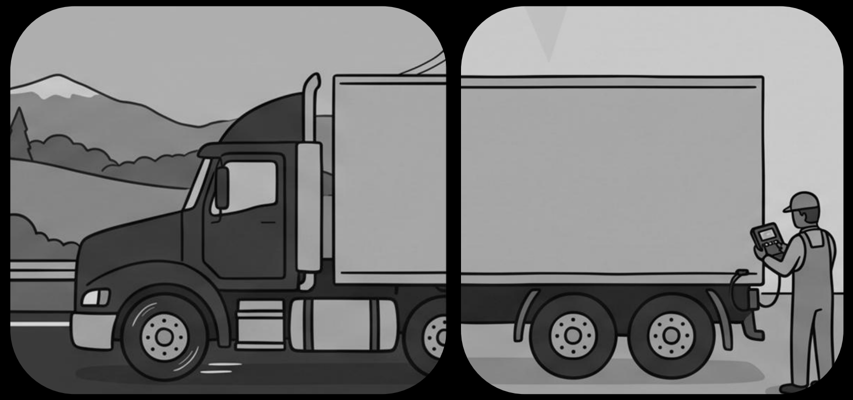
- Tractor brakes are wirelessly accessible via CVE-2022-26131.
- Range is equipment-dependent. Most susceptible is tankers.
- E.g. wireless write to tanker J2497 segment from ~15 ft (~3m) range, for ~300 USD of gear



Bridging the Trust Boundary



Building on Prior Work

This work	Other's Prior work	Our Prior Work	This Works' Firsts
	Hass et. al. 2016 (Wired): commanding tractor systems via J1708	NMFTA + AIS 2020, NMFTA 2022-2024 extends to Wireless J2497	
This work: Mission Time instead of Diagnostic Time	Chatterjee 2024 exploiting J1939 diagnostics on tractors		• First evidence of safety impact on tractors via (wireless) J2497. • First reverse-engineering of a safety recall's firmware update. • First evidence of silent security patching in the motor vehicles.
This work: More in Silent Patches	2019 DEVCORE, PAN-OS GlobalProtect … 2020 Hauser, Cisco Security Manager … 2023 Bishop Fox, GoAnywhere MFT … 2025 Rapid7, Connect Secure		

SECTION: THE RECALL

and the update to remediate it.

The 2024 Bendix EC80 Recall(s)

Apr 2020	~2023 , Sep 2024	Oct 2024		Dec 2024	Feb 2025	Oct 2025	Nov 2025 – Jan 2026	July 17th
Affected Units Produced	**Field Reports, Comp-etitor's TSB**	(Initial) Recall, **ID9363** updater released	**The tip** 🎯	One more OEM	Approached Bendix	Aftermarket parts added	In-Motion Security Impact of Recall Confirmed by NMFTA	Recall Status
"ESC and ATC EC-80 ECUs produced between about April 2020…"	"high electrical noise and low Power Line Carrier (PLC) signal strength"	Recalls 24V-790 and 24V-818 for two OEMs "**memory overwrite** effects"	Engineer flags it to NMFTA	Recall 24V-915		Recalls 25E07-3000, -8000, 7000	Reverse Engineering Completed early 2026	24V780000, 25E07-3000, -7000, -8000: **99.04%** 24V818000: **81.45%** 24V915000: **70.32%**

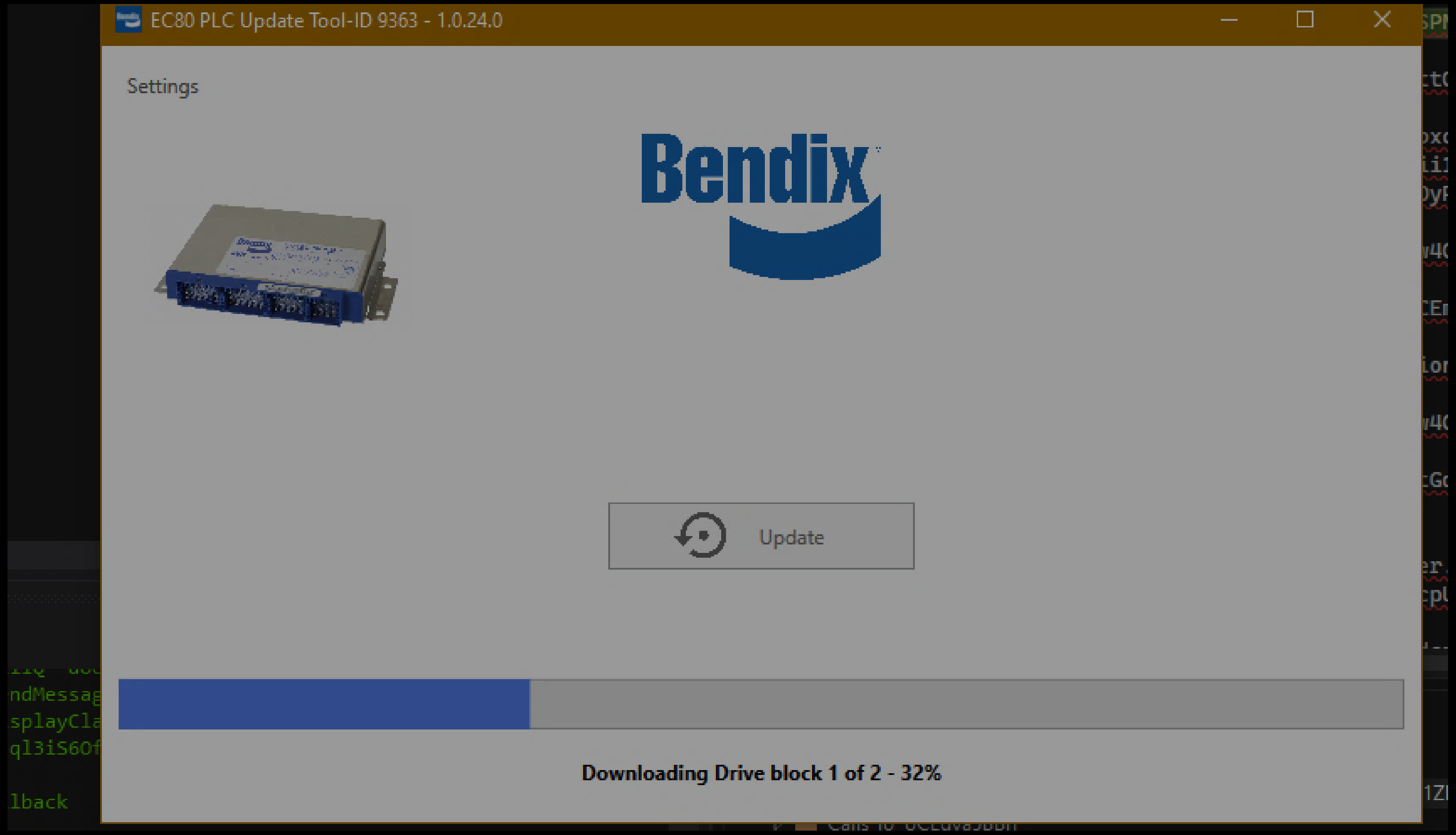
*OEM Original Equipment Manufacturer (they make the trucks)

The Recall's Remediation

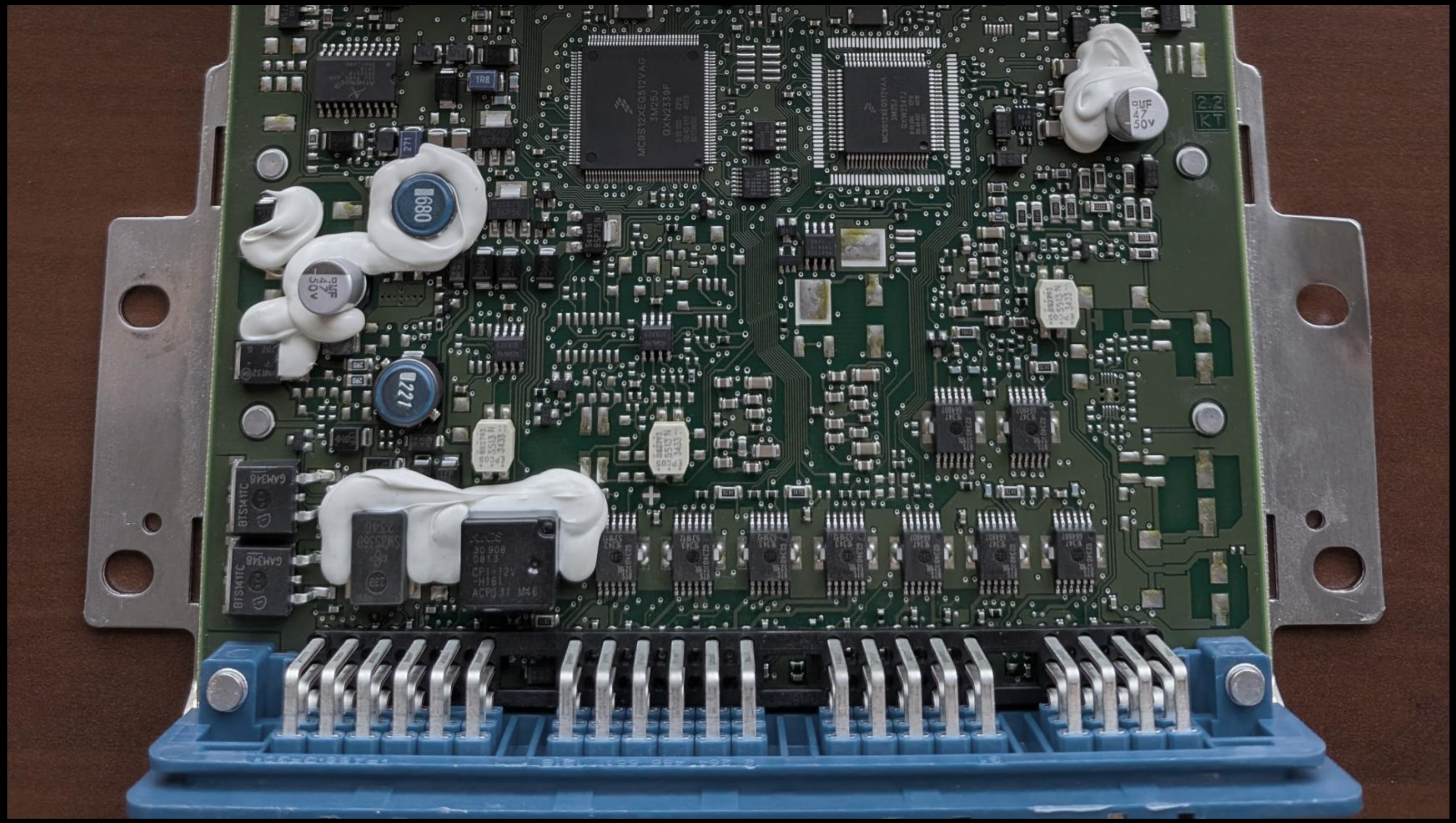
Bendix released ID9363 to remediate all ECUs affected by the recall.

We obtained the executable and used it to update our own ECUs (one for each OEM recall)

Firmware updater software (Windows exe) 'ID9363'

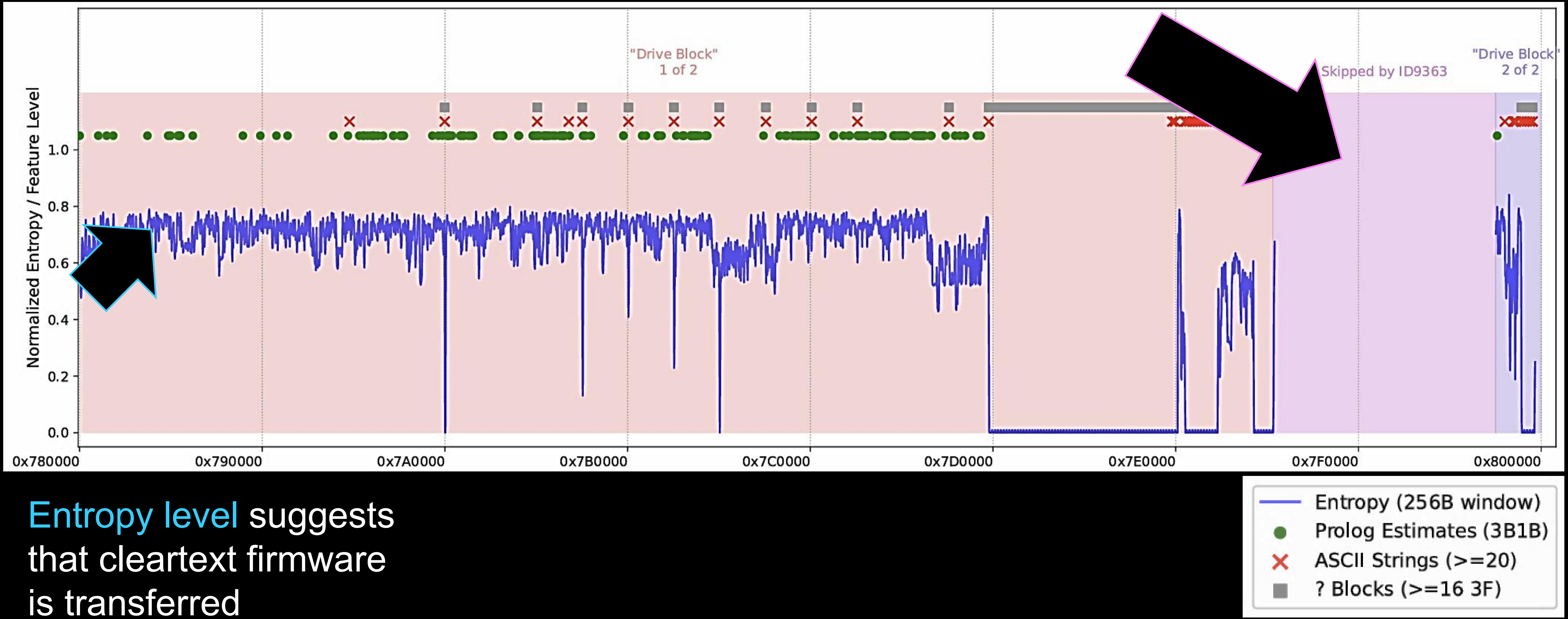


Unified Diagnostic Services (UDS) transfer over J1939 to one of the two S12X MCUs inside the Bendix EC-80 ABS / stability controller



Inside the ID9363 Update

Skipped region is very likely the bootloader hosting the ECU-side transfer



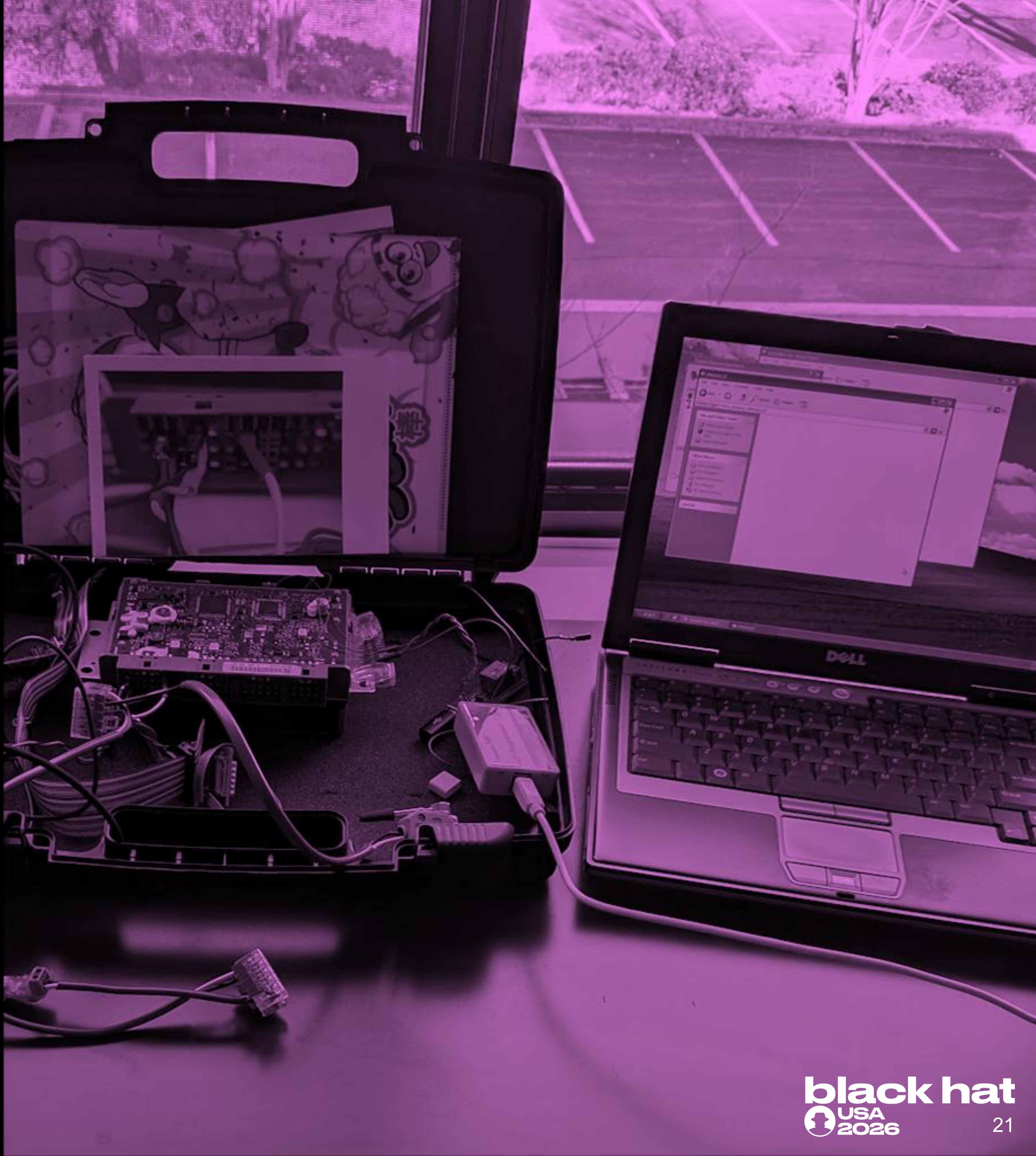
Entropy level suggests that cleartext firmware is transferred

SECTION:

REVERSING & BINDIFFING THE S12X FIRMWARE

From chip to callgraphs.

Extracting Firmware via BDM Interface

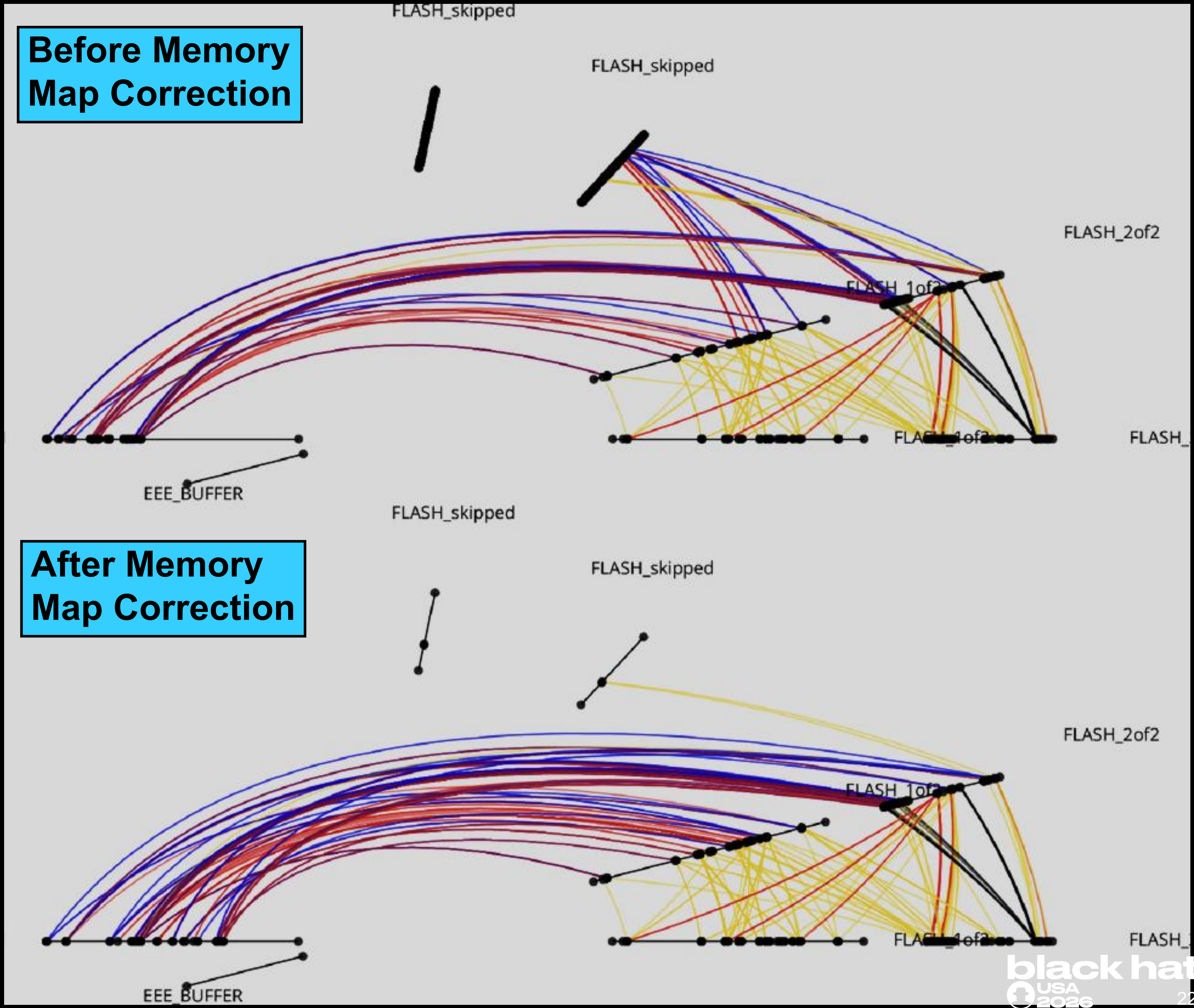


*BDM (Background Debug Module)

Beating Banked Memory (etc.)

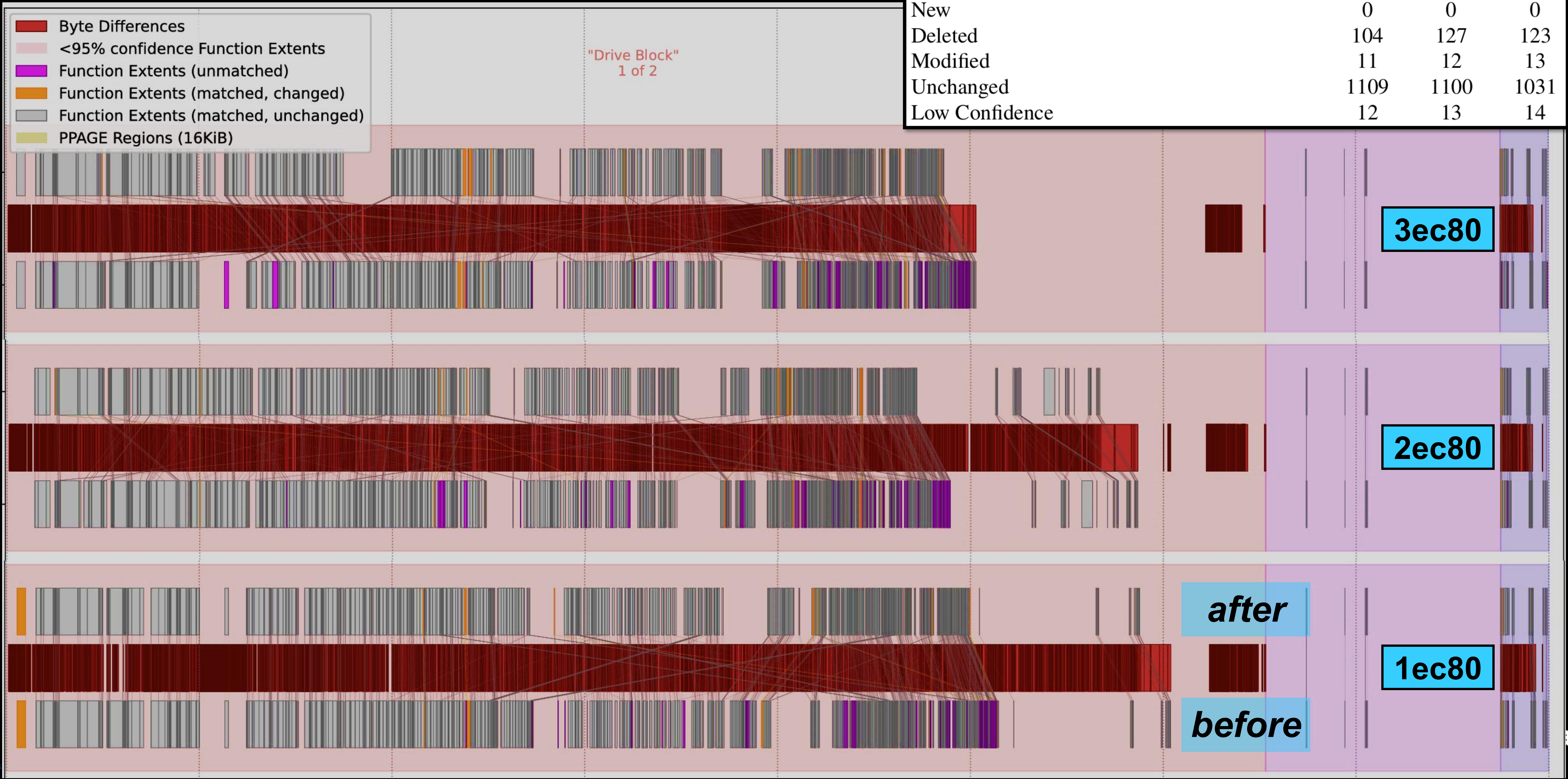
- Making duplicates of data in (banked) segments
- Function-pointer discovery
- PID handlers table parsing
- Basic concolic analysis

*PID: signal ‘types’



Automated QBinDiff Results

	1ec80	2ec80	3ec80
New	0	0	0
Deleted	104	127	123
Modified	11	12	13
Unchanged	1109	1100	1031
Low Confidence	12	13	14

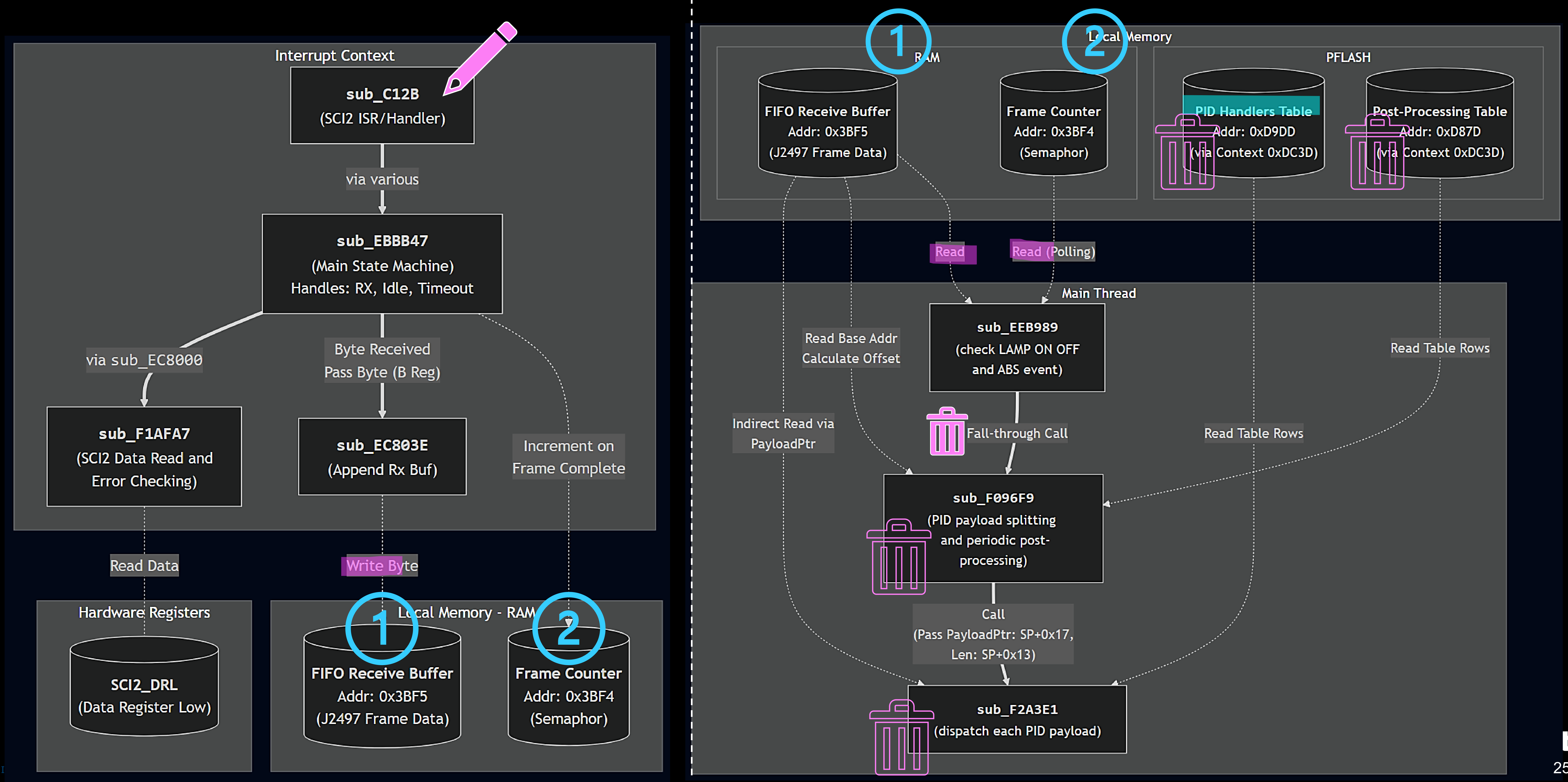


SECTION:

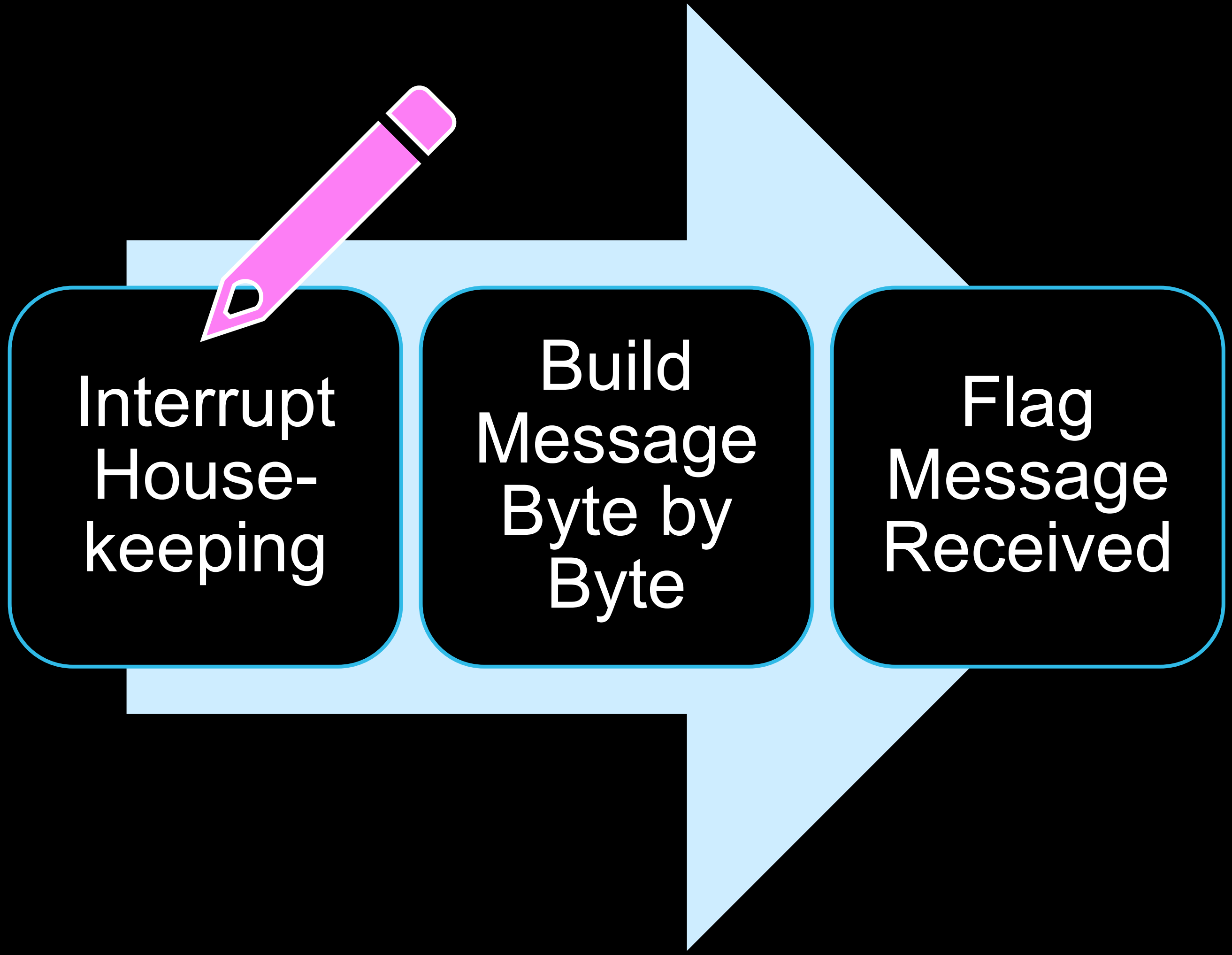
BINDIFF RESULTS & DELETED J2497 ATTACK SURFACE

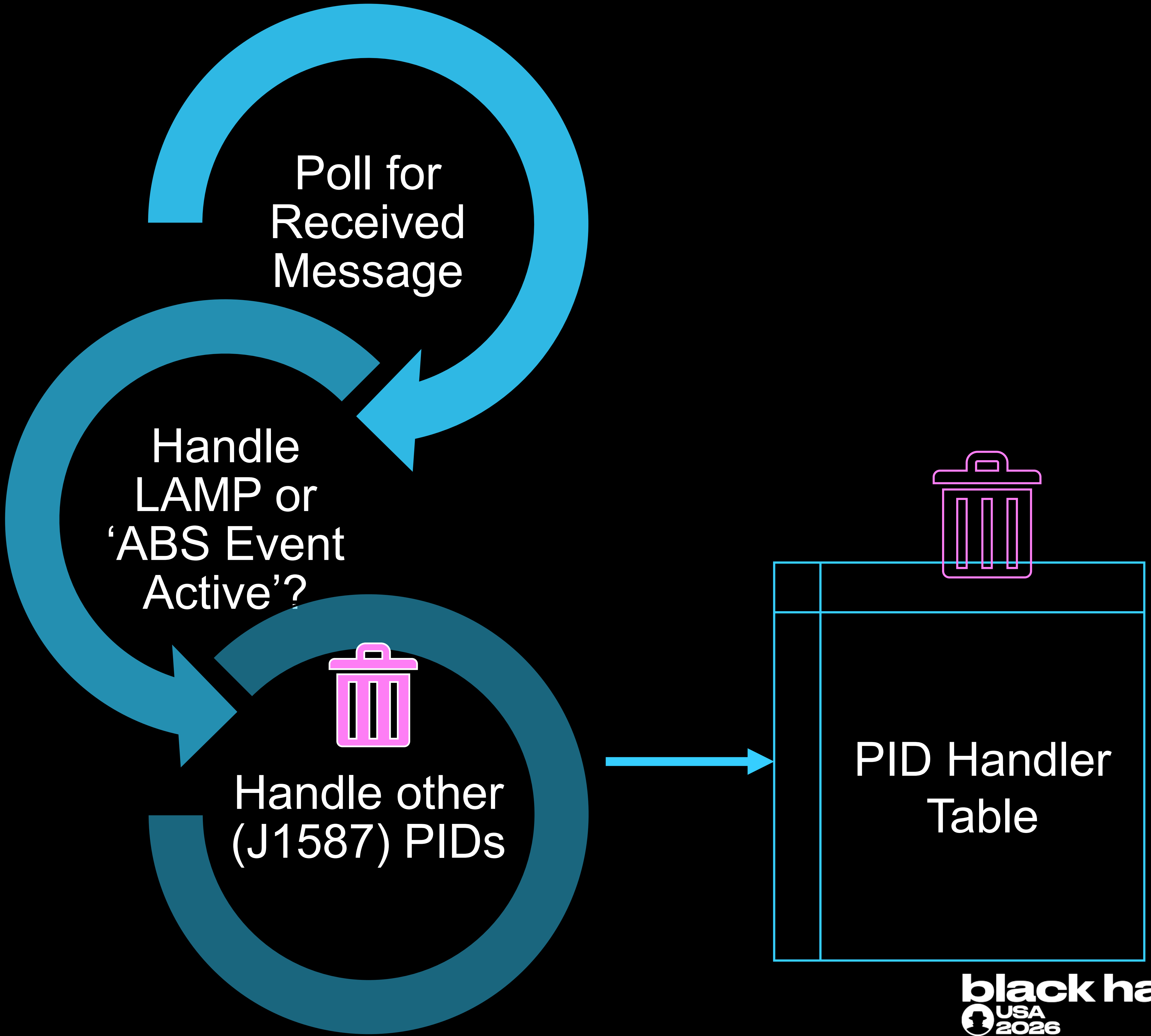
What the ID9363 updater changed.

J2497 Receive Path Changes



J2497 Receive Path Changes





PID Handlers Removed

BYTES MATCH	J1587 STANDARD PID DESCRIPTION	DATA INPUT SIZE
	PID request handlers group	
0031 / 803188	Request for 0x31 ABS Control Status	one byte data (broadcast req) / zero bytes data (unicast req)
	… (13 more request handlers)	
	Streamed PID handlers group	
2A	42 Pressure Switch Status	1 byte data
	… (6 more)	
C2	194 Transmitter System Diagnostic Code and Occurrence Count Table	variable bytes data; byte count followed by sets of diagnostic data
C3	195 Diagnostic Data Request / Clear	3 bytes data
C7	199 Traction Control Disable State	variable length data; byte count, state flags, ASCII access code of 0–15 characters selected by the manufacturer
D1	209 ABS Control Status, Trailer	variable bytes data; up to 3 × 5 bytes for 5 trailers, byte count followed by status bytes
ED	237 Vehicle Identification Number	variable bytes data; byte count followed by VIN ASCII
F5	245 Total Vehicle Distance	4 bytes data
F7	247 Total Engine Hours	4 bytes data
FE88C5 / FE88C6	254 Proprietary DLE	(only 0xC5 and 0xC6)
FF73	115 Trailer Pneumatic Supply Line Pressure	one byte data
FF7B	123 Door Status	one byte data

SECTION:

VULNERABILITIES & DEMOS

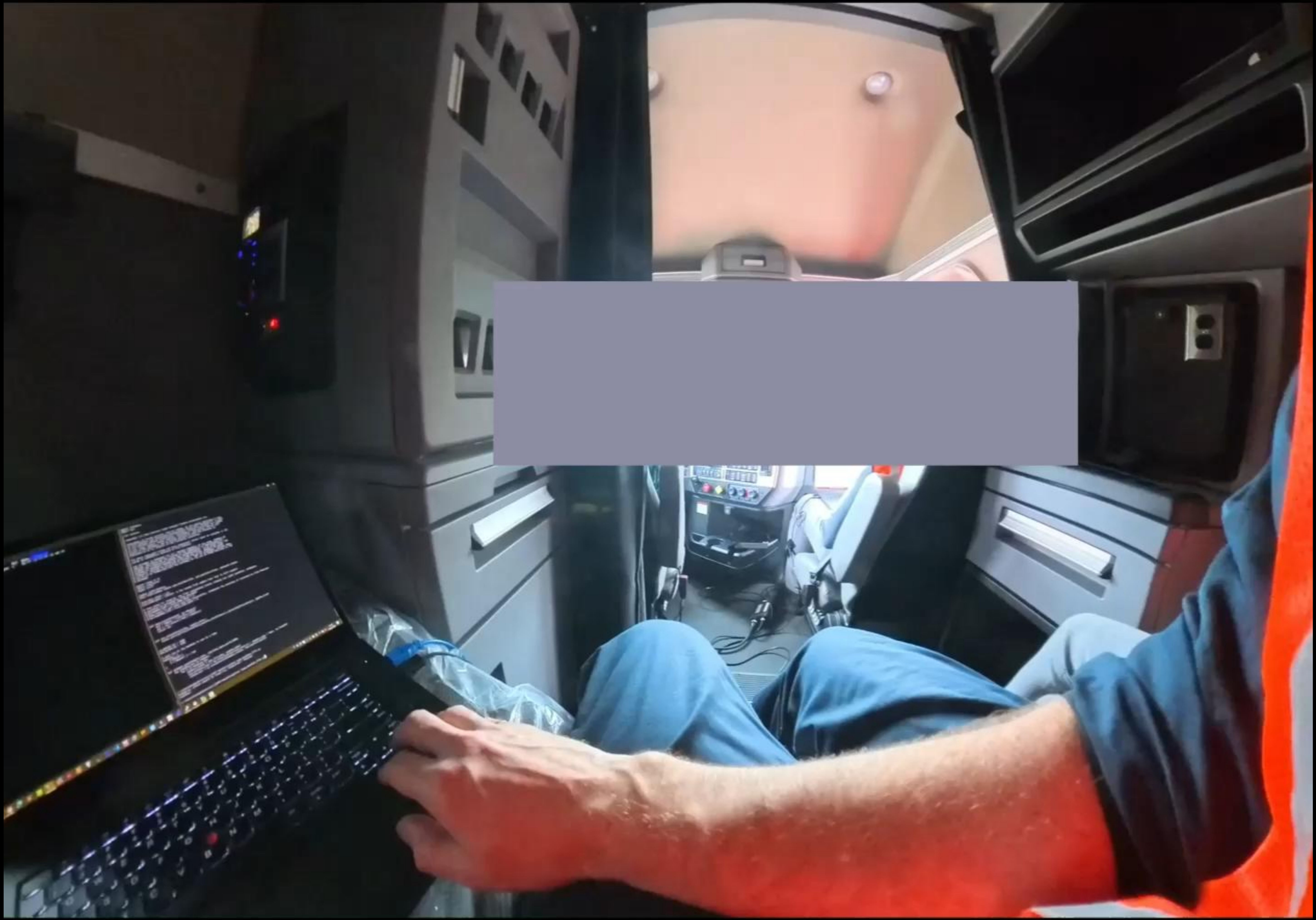
What lived in the deleted code.

Vulns in the Deleted Code

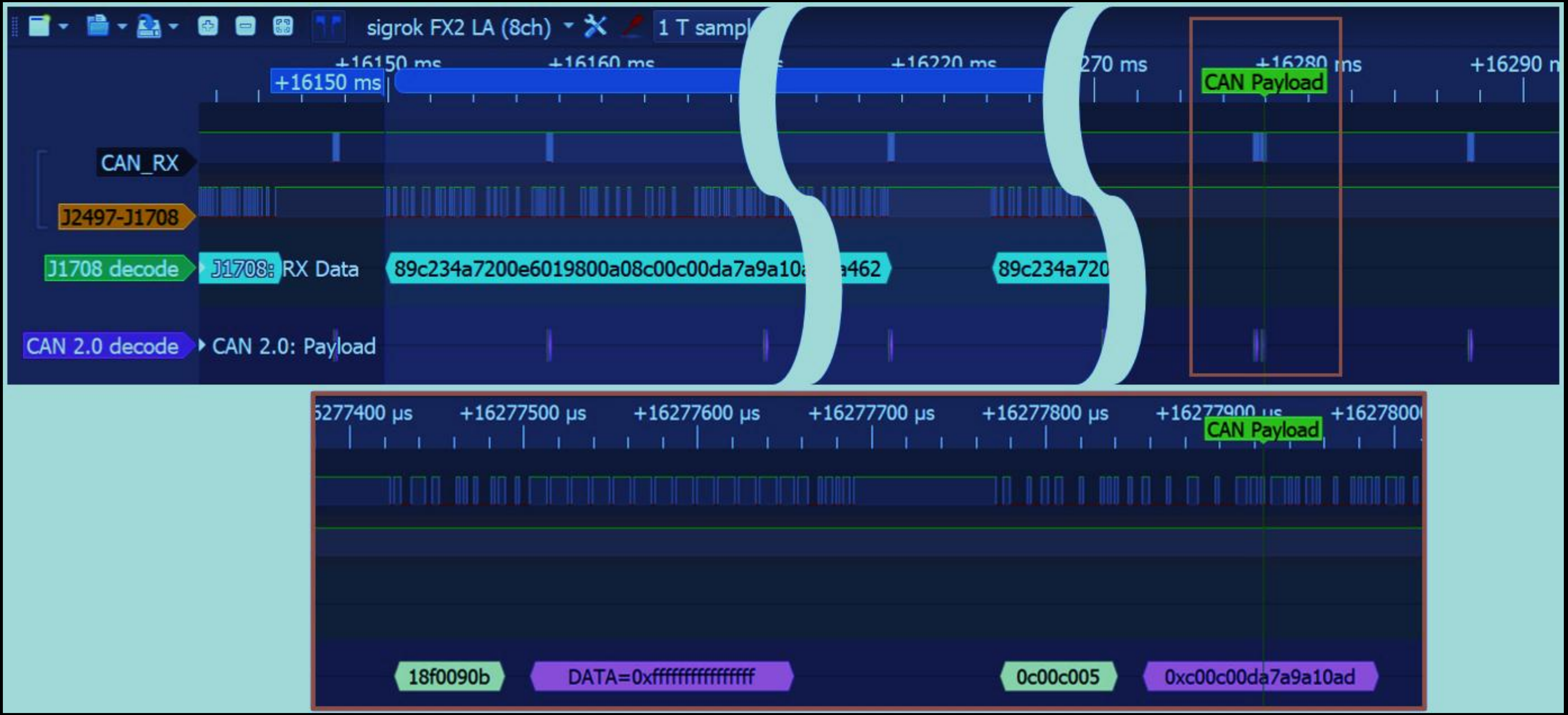
COMPONENT	VULNERABILTIY	IMPACT
PID 0xC2	Buffer Overflow	DoS (Verified), RCE (Verified)
PID 0xED	Unbounded Copy	DoS (Verified), RCE (Theoretical)
PID 0xC7	Hardcoded Credential	Auth Bypass (Verified)
SCI2 EDGE	OOB Write	DoS (Theoretical), RCE (Theoretical)

Demos: DoS on the Track, RCE on the Bench


Control: no attack signal



RCE



sigrok FX2 LA (8ch)
1 T sampl
+16150 ms
+16160 ms
+16220 ms
270 ms
+16280 ms
+16290 m
CAN Payload
CAN_RX
J2497-J1708
J1708 decode
J1708: RX Data
89c234a7200e6019800a08c00c00da7a9a10
a462
89c234a720
CAN 2.0 decode
CAN 2.0: Payload
5277400 µs
+16277500 µs
+16277600 µs
+16277700 µs
+16277800 µs
+16277900 µs
+1627800
CAN Payload
18f0090b
DATA=0xffffffffffffffff
0c00c005
0xc00c00da7a9a10ad

Exploitation/Abuse Mechanisms

VULNERABILITY	METHOD (Verified Only)
PID 0xC2 – Buffer Overflow	**DOS**: Single Message; **RCE**: Uninterrupted Message Sequence
PID 0xED – Unbounded Copy	**DOS**: Repeated Message
PID 0xC7 – Hardcoded Credential	Single Message
SCI2 EDGE – OOB Write 'Race Condition'	**DOS**: Custom J2497 Signals

SECTION: CONCLUSIONS & TAKEAWAYS

What ID9363 really was.

ID9363 is Also a Security Patch

- Removes unauthenticated memory corruptions
  (both DoS and RCE)
- Removes a hardcoded-credential

- Reachable wirelessly
  from various distances (equipment-dependent)
- Reachable by compromised trailer telematics

On 'Random Noise'

EDGE: plausible as a noise triggered bug

Now Aligned With Best Practices

	Guidance	AFTER ID9363
ATA TMC PP 2024-3	Restrict tractor J2497 to lamp messages	✅
CISA ICSA-25-021-03	Minimize reachable attack surface	✅
SAE J2497 (2026)	Receive ABS warning lamp **only**	✅

The Recall Effort is Commendable

Best Practice Documents:

- ISO/IEC 29147 Information Technology– Security Techniques– Vulnerability Disclosure
- CISA “*Shifting the Balance of Cybersecurity Risk: Principles and Approaches for Security-by-Design and-Default*”

Security patches should be communicated to users so that they can make their own informed risk calculations.

Acknowledgements

- This work would not be possible without the support of the member fleets of the NMFTA Inc.
- AIS and AFRL for access to their Class 8 vehicle multiple times during this research.
- Hannah Silva and Jesse Norton for sharing S12X development materials and EC80 experience.
- Chris York for the trailhead.
- Jonatan Mars for critical support at a critical time.
- Many industry engineers for their support of this research– the rest of whom would prefer not to be named.

- We used various large language models (Gemini 2, 2.5 pro, 3-pro) for tasks such as: drafting Python code and Mermaid diagrams.
- We acknowledge the open-source tools used: Zynamics BinDiff, Quarkslab QBinDiff, Quokka, Python 3, python-can, py-hv-networks, RP1210 python, Osmocom FL2K

THANK YOU

White paper:

https://i.blackhat.com/BH-USA-26/Presentations/BHUS26-Gardiner-Tractor_ECU_RE-WP.pdf

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USA
2026